

Riding a suitcase?

Coolpeds has built an electric scooter that zooms around at 12.4 mph, and also works as a suitcase <http://dgit.in/Coolpeds>

Free Office 365

Colleges that have purchased Office for faculty, makes the students eligible for a free service of Office 360. <http://dgit.in/FreeOff365>

cast have access to. However, with compatibility across all streaming protocols you come to the realisation that there are workarounds for these which means you don't feel disadvantaged. There are apps for all mobile platforms and one for the desktop as well. Even the hardware is the same across all dongles so picking an EZCast dongle comes down to whether it has an external antenna coupled with the accessories that each manufacturer is giving away with them.

The Stones

The not so inconspicuous cousin of the streaming sticks are the boxes which outclass the sticks when it comes to features and processing power. We ended up with two products from Minix, a Hong Kong based company that makes Android based TV hubs. Then we had the iRevo, an Indian brand that launched a product without any fanfare last year. And lastly, we have the Raspberry Pi 2 which we threw into the mix just to see how well it fared. Since this was something we built, it would be unfair to score the Raspberry Pi against the others which is why it was put in purely for representational purposes. We missed out on the Fire TV, Google TV and Apple TV in the boxes category but it didn't seem to be a poor comparison given the hardware we had at hand.

iRevo SMART TV has something unique that none of the other contenders have. The UI they've built though slow at times has Indian content which was relevant to us since it added up to a higher ROI (Return On Investment). The remote has a built in Gyroscope which works wonderfully across the interface. The UI seemed as if it were in a prototype phase and there

were advertisements all over the place. Watching adverts in between TV shows and movies is bothersome and it isn't an element that we wanted to see on the interface as well. It's a good device that could have been a lot better if they'd included more than just two USB ports on the rear. The firmware is Android

KODI which by far is the best option. Since you do have access to the underlying Android OS you can fiddle with all sorts of settings and are not restricted as was the case when we checked out the Fire TV a few months back. Minix doesn't provide an airmouse with the device (just an IR remote) but you can buy



Make your own streaming device with Raspberry Pi 2 and KODI

based so you can chuck all of that and run whichever apps you fancy.

Minix's Neo X7 and the more recent X8-H Plus turned out to be completely out of the ordinary. The hardware is way ahead of the competition but the cost seems to be a hindrance to the device gaining mass appeal. It has a card reader, OTG port, three USB ports, a detachable antenna, ethernet port and provision for analog and digital A/V output. It's build quality is by far the best we have seen in this test. The analog A/V output is obtained from the audio jack which is a CVBS port once you force the firmware to push analog signals. It runs on Android with a custom skin on it and the best thing about it would be that it has an installation of KODI(XBMC is now known as KODI) on it. So you have a choice between sticking to the Minix UI or switching to vanilla android or running

Neo A2 separately which turns it into a console of sorts. The remote being an airmouse has a gyroscope, microphone and a qwerty keyboard on the rear.

The Barebones

Raspberry Pi 2 Model B was thrown in just to see how a plain KODI installation plays against the rest of the competition. This, however, requires some work but you have a lot more flexibility at the end of it just like any Linux installation. But if that doesn't do it for you, then when Windows 10 rolls out each RPi 2 gets a lean version of the OS for free. We attached a TP-Link TL-WN725N 150 Mbps Wi-Fi adapter to the RPi 2 and used a 32 GB Sandisk Ultra MicroSD card for the OS image. Installing Piracast makes the RPi 2 capable of Miracast but this plugin is still being perfected so performance isn't great at the moment. Moreover, Piracast only works with the aforementioned TP-

Link dongle as that's the only one the programmer got to play with. KODI does have Airplay support straight out of the box and if you were to spend a little more to get the Skype SDK for linux then you can run Skype as well. So if it were upto us nerds to look at all that we have at our disposal, it would be very difficult to pass on the RPi as a complete package.

Build Quality

Not much can be said about the dongle which were designed in a similar fashion. EZCast has a hardware specification for their dongles and all manufacturers have done nothing to fiddle around with the hardware except for Cubetek and Ematic who have attached an external antenna on the device. So internally the dongles are exactly the same. The new Chromecast devices are known to heat up quite a bit so you might end up seeing your Chromecast shut down due to overheating an awful lot many times but YouTube does have plenty of videos where folks have modified heatsinks and antenna hacks to improve the device. Except for the heating issue, the device is quite well built. Among the boxes, the iRevo was very lightweight and after opening the box we noticed that the PCB occupies a very small area inside the device. This does mean there is a lot more airflow but we prefer a compact device. Minix's devices were all built the way we'd like other devices to be. The boxes were weighed down with heavy base plates and the X8-H Plus has rubber feet. Both of these combined make it sturdy and stay where you place them. Add to this the fact that the antenna is detachable and you can extend/swap it which makes it easy to hide away.